

CNN Architecture: Conv → Pool → Dense

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

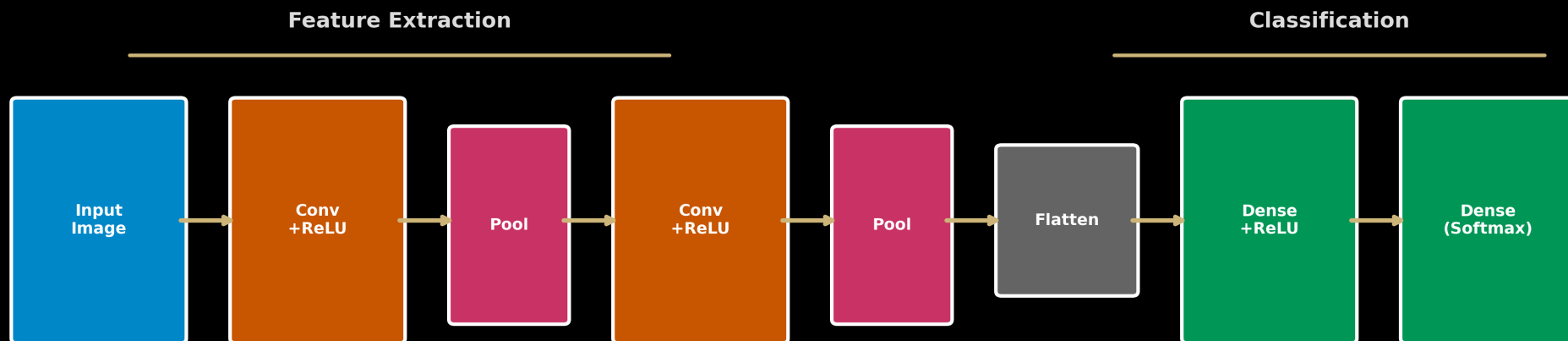
Contents of This Video

In this video, we will cover:

- The standard CNN pipeline
- Feature extraction layers (conv + pool)
- Classification layers (dense)
- A complete example: LeNet-5
- Tracing dimensions through the network

The Standard CNN Pattern

Standard CNN Architecture



Two stages:

1. **Feature extraction:** Conv + ReLU + Pool (repeated)
2. **Classification:** Flatten + Dense layers

Feature Extraction Stage

What happens at each conv + pool block:

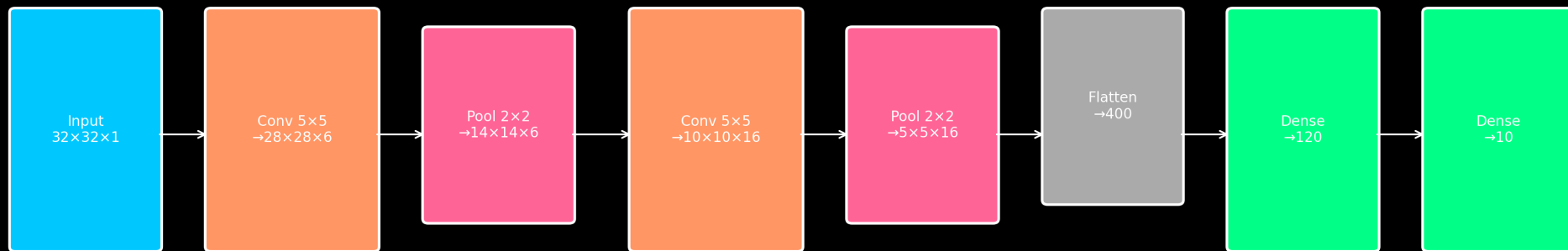
Layer	Operation	Effect
Conv	Apply filters	Detect features
ReLU	Non-linearity	Enable complex patterns
Pool	Downsample	Reduce size, add invariance

Each block:

- Doubles (roughly) the number of filters
- Halves the spatial dimensions
- Builds more abstract features

Example: LeNet-5

LeNet-5: A Classic CNN (1998)



Parameters: ~60,000 (tiny by modern standards!)

Tracing Dimensions: Step by Step

Input: $32 \times 32 \times 1$ (grayscale image)

Conv1 (6 filters, 5×5): $32 - 5 + 1 = 28 \times 28 \times 6$

Pool1 (2×2 , stride 2): $28 / 2 = 14 \times 14 \times 6$

Conv2 (16 filters, 5×5): $14 - 5 + 1 = 10 \times 10 \times 16$

Pool2 (2×2 , stride 2): $10 / 2 = 5 \times 5 \times 16$

Flatten: $5 \times 5 \times 16 = 400$

Dense1: 120 neurons

Dense2 (output): 10 neurons (digits 0-9)

LeNet Architecture Summary

Layer Type	Configuration	Output Shape
Conv2D	6 filters, 5×5	28×28×6
MaxPool	2×2	14×14×6
Conv2D	16 filters, 5×5	10×10×16
MaxPool	2×2	5×5×16
Flatten	—	400
Dense	120 neurons + ReLU	120
Dense	10 neurons + Softmax	10

Total parameters: ~60,000

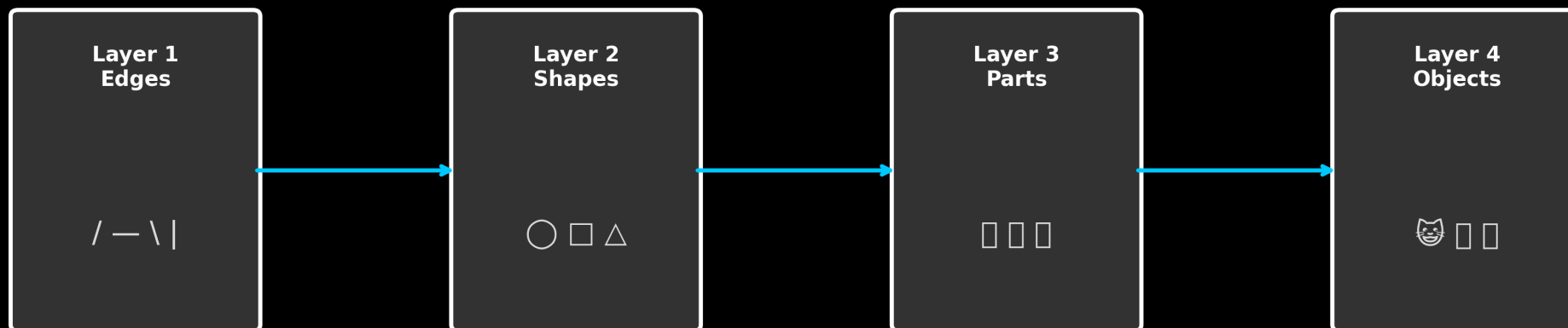
Hierarchical Feature Learning

Early layers: Edges, simple textures

Middle layers: Shapes, parts

Late layers: Objects, faces

CNNs Learn Hierarchical Features



What We've Covered

In this video, we've learned:

- CNNs have two stages: feature extraction and classification
- Feature extraction: repeated Conv + ReLU + Pool blocks
- Classification: Flatten + Dense layers
- LeNet-5: the original CNN pattern (1998)
- How to trace dimensions through a network
- CNNs learn hierarchical features automatically